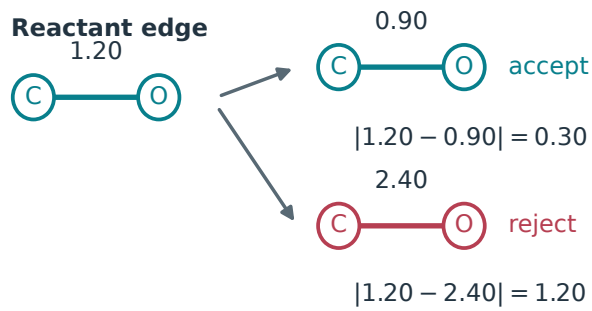


a Weighted matching

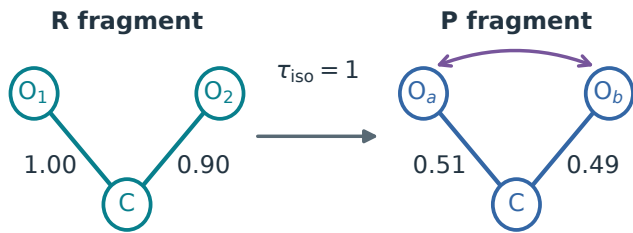
R-P weight compatibility controls admissible placements.



Example: $\tau_{\text{iso}} = 1.0$; edge floor $b = 0.2$.

c Compress with automorphisms

Generators preserve matching tests, not numerical weights.

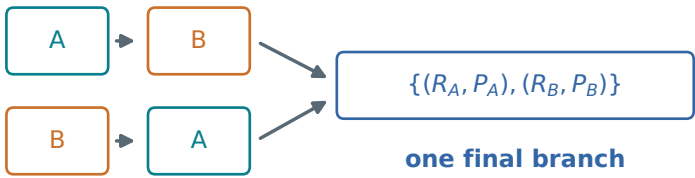


Generator: $g = (O_a O_b)$

Both O assignments pass the matching tests.
The group encodes coupled atom moves.

e Deduplicate final fragment pairs

Growth order disappears; internal alternatives remain.

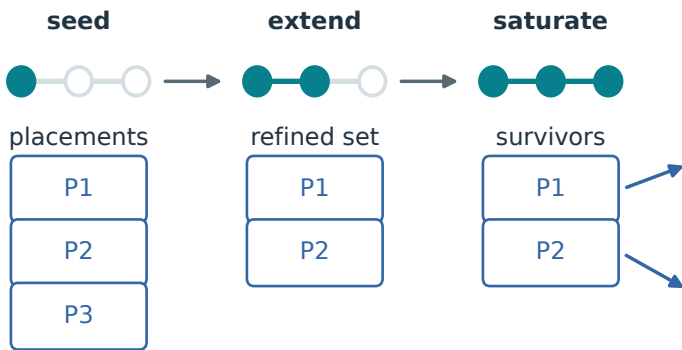


Branch contents: $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$

Share identical records; retain different constraints, actions, and witnesses.

b Grow, refine, then branch

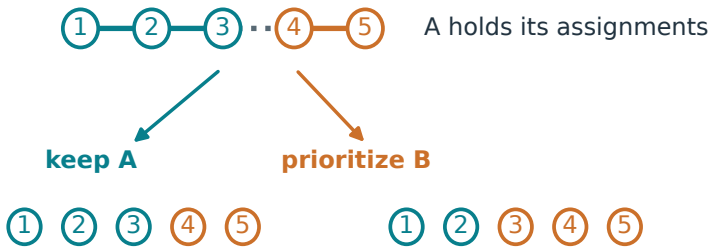
Grow even when only one placement remains.



Continue remaining atoms under each placement.

d Let neighboring fragments compete

Keep the original branch; add a challenger continuation.



Rematch B at the contact; release conflicts, anchor unaffected atoms, and regrow gaps.

f Decode distinct bond-event classes

Project allowed mappings onto signed bond changes.

O assignment in (c)	ΔW	events
$O_1 \mapsto O_a, O_2 \mapsto O_b$	-0.49, -0.41	0
$O_1 \mapsto O_b, O_2 \mapsto O_a$	-0.51, -0.39	1

At $\delta = 0.5$: the second mapping weakens C-O₁.



Repeat until the event window is exhausted.